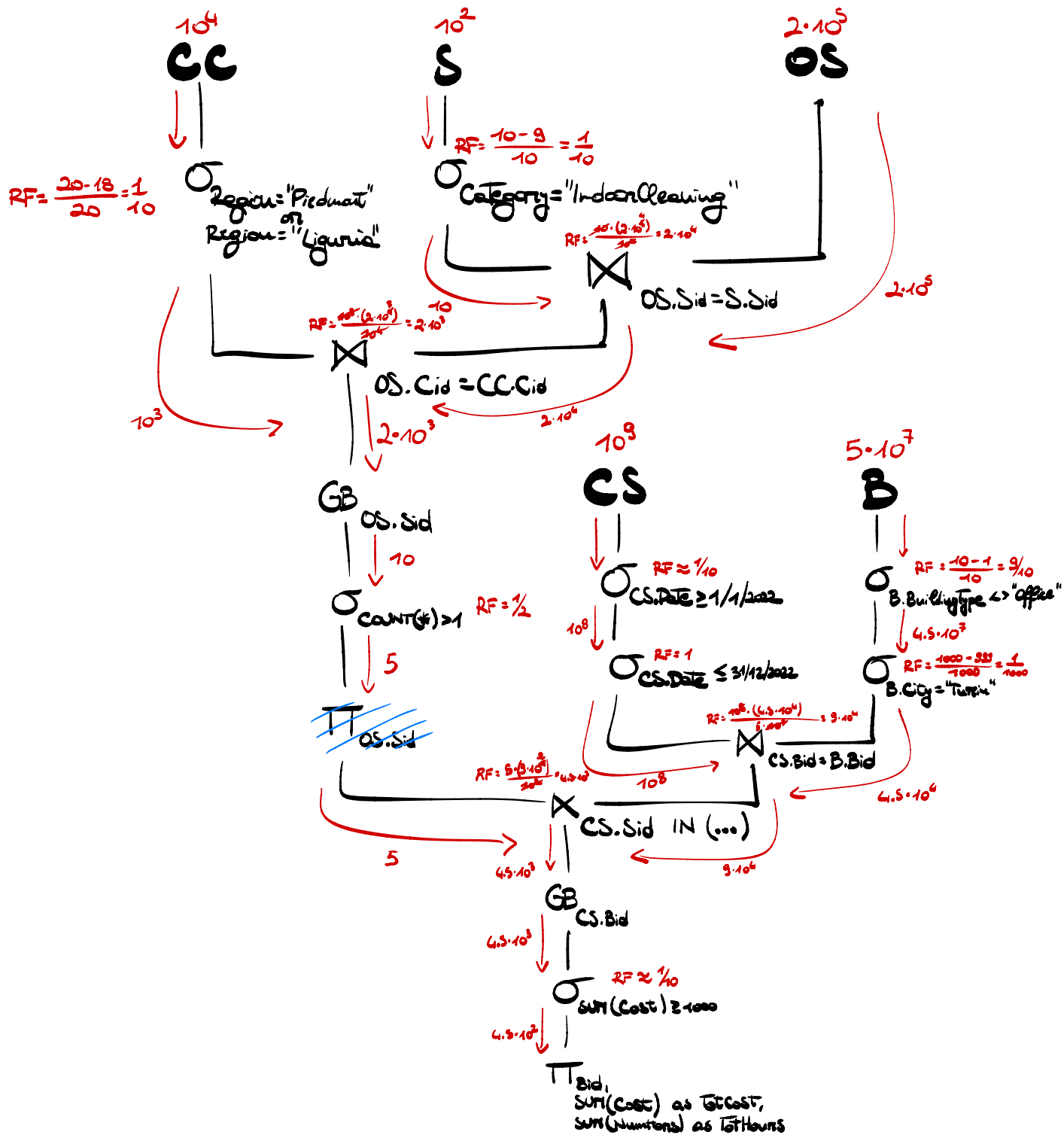
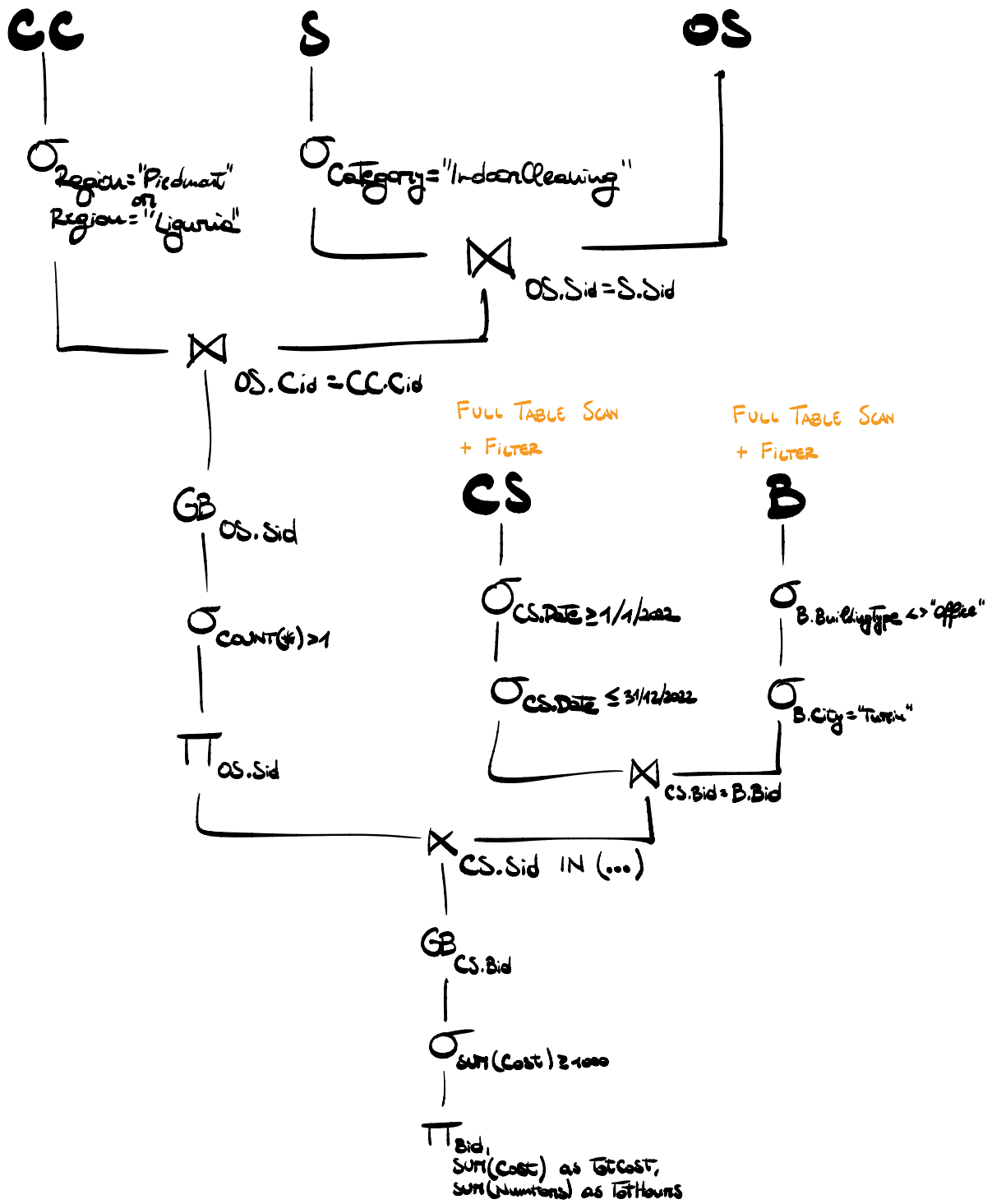


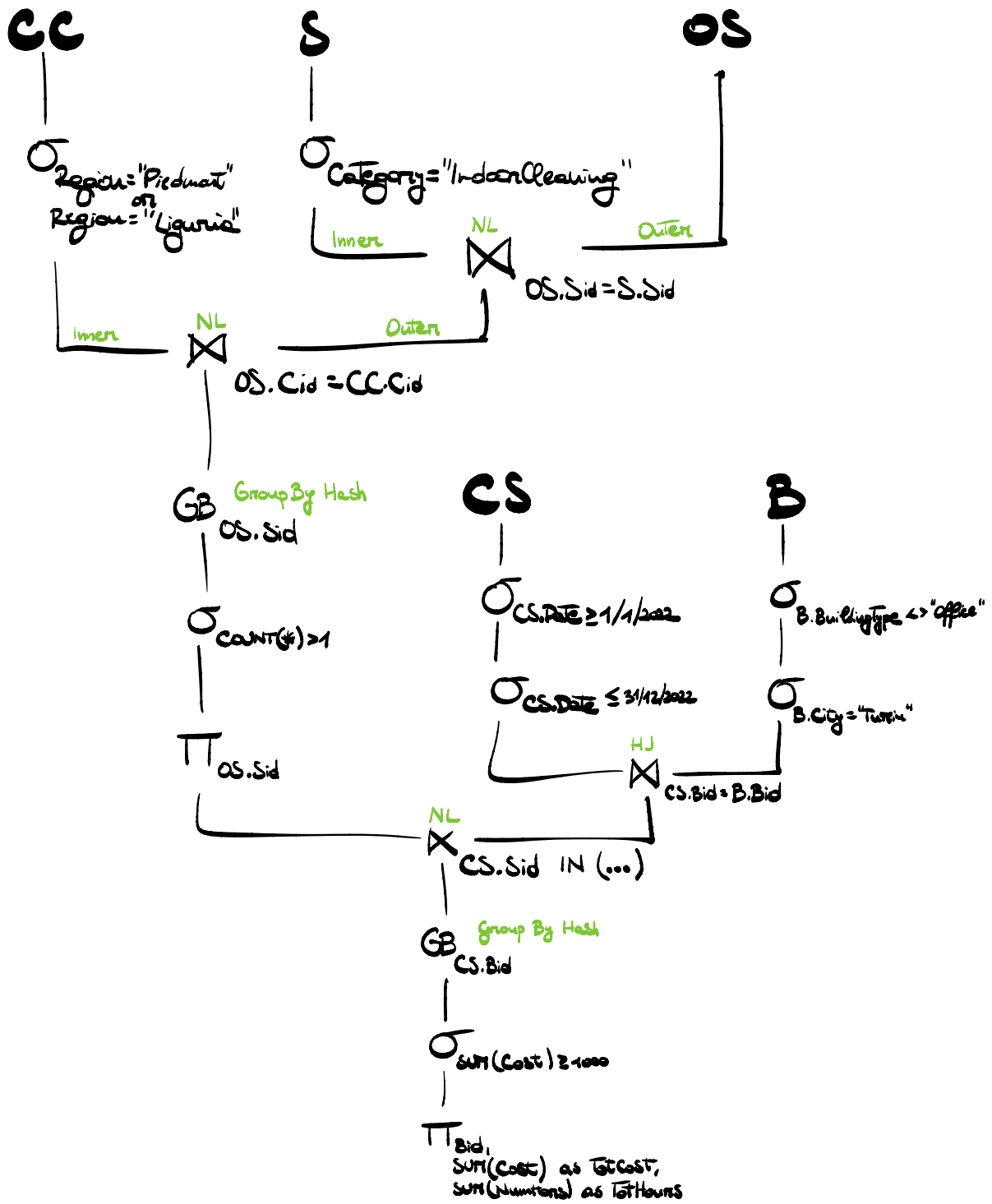


FULL TABLE SCAN
+ FILTER

FULL TABLE SCAN
+ FILTER

FULL TABLE SCAN





Index Range Scan + Access by RowID
Secondary index: Hash on Region

CC

$\sigma_{S=1/10}$
 $\sigma_{\text{Region}="Piedmont"}$
 $\sigma_{\text{Region}="Liguria"}$

Index Range Scan + Access by RowID
Secondary index: Hash on Category

S

$\sigma_{S=1/10}$
 $\sigma_{\text{Category}="IndoorCleaning"}$

Full scan Index
Primary index: B+Tree on Cid, Sid

OS

$OS.Sid = S.Sid$

$OS.Cid = CC.Cid$

GB

$OS.Sid$

$\sigma_{COUNT(*) > 1}$

$\Pi_{OS.Sid}$

Index Range Scan + Access by RowID
Secondary index: B+Tree on Date

CS

$\sigma_{S=1/10}$
 $\sigma_{CS.Date \geq 1/1/2022}$

$\sigma_{S=1}$
 $\sigma_{CS.Date \leq 31/12/2022}$

Index Range Scan + Access by RowID
Secondary index: Hash on BuildingType

B

$\sigma_{S=3/10}$
 $\sigma_{B.BuildingType \neq "office"}$

$\sigma_{S=1/1000}$
 $\sigma_{B.City="Turin"}$

$CS.Bid = B.Bid$

$CS.Sid \text{ IN } (...)$

GB

$CS.Bid$

$\sigma_{SUM(Cost) \geq 1000}$

$\Pi_{Bid, SUM(Cost) \text{ as TotCost, SUM(NumHours) as TotHours}$